

tf.compat.v1.reduce_logsumexp Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/reduce_logsumexp

*Computes $\log(\text{sum}(\exp(\text{elements across dimensions of a tensor})))$.
(deprecated arguments)*

Function Signatures

```
tf.compat.v1.reduce_logsumexp( input_tensor, axis=None,  
keepdims=None, name=None, reduction_indices=None,  
keep_dims=None )  
(keep_dims)
```

Code Examples

```
tf.compat.v1.reduce_logsumexp(  
    input_tensor,  
    axis=None,  
    keepdims=None,  
    name=None,  
    reduction_indices=None,  
    keep_dims=None  
)
```

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tf.compat.v1.reduce_logsumexp(  
    input_tensor,  
    axis=None,  
    keepdims=None,  
    name=None,  
    reduction_indices=None,  
    keep_dims=None  
)
```

```
x = tf.constant([[0., 0., 0.], [0., 0., 0.]])  
tf.reduce_logsumexp(x) # log(6)  
tf.reduce_logsumexp(x, 0) # [log(2), log(2), log(2)]  
tf.reduce_logsumexp(x, 1) # [log(3), log(3)]  
tf.reduce_logsumexp(x, 1, keepdims=True) # [[log(3)], [log(3)]]  
tf.reduce_logsumexp(x, [0, 1]) # log(6)
```

```
x = tf.constant([[0., 0., 0.], [0., 0., 0.]])  
tf.reduce_logsumexp(x) # log(6)  
tf.reduce_logsumexp(x, 0) # [log(2), log(2), log(2)]  
tf.reduce_logsumexp(x, 1) # [log(3), log(3)]  
tf.reduce_logsumexp(x, 1, keepdims=True) # [[log(3)], [log(3)]]  
tf.reduce_logsumexp(x, [0, 1]) # log(6)
```

Documentation

Computes $\log(\sum(\exp(\text{elements across dimensions of a tensor})))$. (deprecated arguments)

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.math.reduce_logsumexp`

Used in the notebooks

- [Fitting Dirichlet Process Mixture Model Using Preconditioned Stochastic Gradient Langevin Dynamics](#)

Reduces `input_tensor` along the dimensions given in `axis`.

Unless `keepdims` is true, the rank of the tensor is reduced by 1 for each of the entries in `axis`, which must be unique. If `keepdims` is true, the reduced dimensions are retained with length 1.

If `axis` has no entries, all dimensions are reduced, and a tensor with a single element is returned.

This function is more numerically stable than $\log(\sum(\exp(\text{input})))$. It avoids overflows caused by taking the exp of large inputs and underflows caused by taking the log of small inputs.

For example:

Returns